Sri Sai Vidya Vikas Shikshana Samithi ®
SAI VIDYA INSTITUTE OF TECHNOLOGY
Approved by AICTE, New Delhi, Affiliated to VTU, Recognized by Govt. of Karnataka.
Accredited by NBA, New Delhi (CSE (Data Science)), NAAC – "A" Grade
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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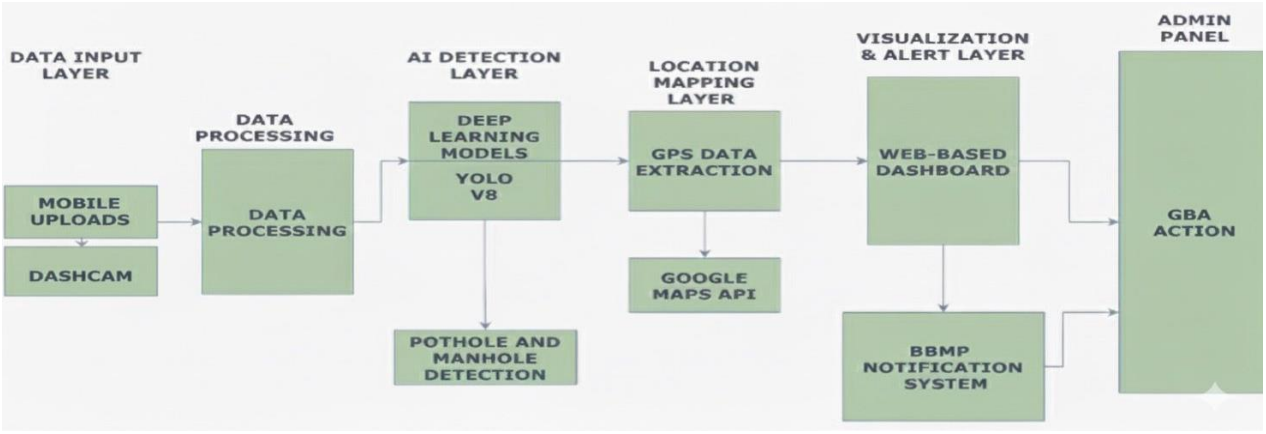
Project Synopsis

Title: Smart Road Infrastructure: AI-Powered Pothole Detection & Alert

Introduction: Bangalore's pothole-damaged roads create major safety risks and traffic problems, and because the current citizen-based reporting system is slow and inefficient, this project proposes an AI-driven solution that analyzes dashcam video using deep learning to detect potholes and send real-time alerts to GBA for quick action.

Problem Statement: Bangalore's roads suffer from frequent potholes that pose significant safety risks to commuters and pedestrians, and the absence of a real-time automated system to detect and report these issues leads to delays in maintenance and increased chances of accidents.

Proposed System:



Objectives:

- **Develop a deep learning-based model** for accurate pothole and open manhole detection using **YOLOv8**.
- **Automate the alert system** to **notify the Greater Bengaluru Authority (GBA)**
- with precise geolocation data, ensuring faster repairs and improved road safety.
- **Utilize multiple data sources** for comprehensive and effective road monitoring such as (video feed from vehicle-mounted dashcams and mobile uploads).
- **Implement a real-time dashboard** for **visualizing, tracking, and analysing detected road issues** efficiently.

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Requirements Specification:

- **Hardware Requirements:**
- **Processor**: Minimum **Intel i5 / Ryzen 5** or equivalent processor.
- **RAM:** At least **8 GB RAM** (16 GB recommended for faster training).
- **Storage: SSD storage** (256 GB or more) for datasets, models, and logs
- **Internet:** Stable connection for downloading models/updates

- **Software Requirements:**
- **Operating System:** Windows 10/11, Linux, or macOS
- **Programming Language:** Python 3.8
- **Frameworks & Libraries:**
- **Ultralytics YOLOv8** (for object detection)
- **PyTorch** (backend for YOLOv8)
- **OpenCV** (for video frame extraction and processing)
- **NumPy, Pandas** (for data handling and analysis)
- **Matplotlib / Seaborn** (for graphs in reports)
- **Development Tools:** VS Code, Jupyter Notebook, Git/GitHub

- **Functional Requirements:**

- **Real-time pothole detection**
- **Video frame extraction**
- **YOLOv8 object detection**
- **Pothole image capture**
- **GPS location mapping**
- **Automated report generation**
- **Email alert system**
- **Web dashboard display**
- **User video/image upload**
- **Database storage of detections**

- **Non-Functional Requirements:**

- **Performance** – The system should process video frames quickly for near real-time detection.

- **Accuracy** – The YOLOv8 model should maintain high detection accuracy for potholes.
- **Scalability** – The system should support multiple users uploading videos simultaneously.
- **Reliability** – The system should consistently detect potholes without frequent failures.
- **Usability** – The dashboard should be simple and easy for users and authorities to operate.
- **Security** – User data, video uploads, and location details must be securely handled.
- **Availability** – The system should be available online for user access at most times.
- **Efficiency** – The system should use computational resources optimally during detection and reporting.

➢ **Decision Fusion & Output:**

- Combines model predictions from video frames.
- Filters false positives using confidence thresholds.
- Selects the most accurate pothole detection per frame.
- Generates final detection results with bounding boxes.
- Maps each confirmed detection with GPS coordinates.
- Produces a summarized pothole report.
- Sends fused results to the dashboard.
- Triggers real-time email alerts to authorities.
- Displays visual and analytical results to the user.

Support for follow-up queries

Team Details:

Sl.No.	USN	Name	Signature
01.	1VA22CD084	Prakyath Gowda P	
02.	1VA22CD007	Anushree P	

Signature of Guide **Signature of Coordinator** **Signature of HOD**